

UNIVERSITY OF GHANA

DCIT 208- SOFTWARE ENGINEERING

INDIVIDUAL ASSIGNMENT 2

CLIENT INTERVIEW EVIDENCE AND REQUIREMENT
TRANSFORMATION MEMO

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Team Name: InnovateX

Project Title: ScentHub

Client: Yussif Kangre Abdulai

Team Repository URL: <https://github.com/selasi07/ScentHub>

Individual Portfolio Repository URL: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-individual-engineering-evidence-portfolio-2026-kellyadoboe1>

Submission Date: 3 July 2026.

AI Disclosure Statement

I used ChatGPT (OpenAI) to help improve the wording, structure, and organisation of this report after completing my own work. I reviewed all generated content, modified it to reflect the ScentHub project, and verified that it accurately represented the client discussion and assignment requirements. I understand that I am responsible for explaining and defending every part of this submission.

Client Discovery Questions

1. Can you explain how you currently record perfume sales and update your inventory after each sale?
2. What steps do you follow when new perfume stock arrives at your shop?
3. What are the biggest challenges you experience with your current manual inventory and sales recording process?
4. Have you ever encountered situations where products were sold even though they were out of stock? If yes, how did you handle them?
5. What information do you normally record for each sale (e.g., perfume name, quantity, selling price, customer details)?
6. What information about each perfume product do you think the system should store to help you manage your business effectively?
7. What improvements would make you consider the ScentHub system successful for your business?
8. Which reports or business information would you like the system to generate for you regularly?
9. Who should be allowed to access the system, and are there any business records or customer information that should only be visible to certain users?
10. What should the system do if a staff member tries to record a sale for a perfume that is out of stock or enters incorrect information?
11. How many people will use the system, and what different roles or responsibilities will they have?
12. Are there any future features or improvements you would like the system to support as your business grows?

Interview Information

Date: Tuesday, 30 June 2026

Time: 1:15 PM – 1:45 PM

Medium: WhatsApp Video Call

Participant Role: Business Owner (Client)

Evidence Type: Notes taken during the discussion. No audio or video recording was made to respect the client's privacy.

Interview Summary

The discussion focused on understanding how the client manages perfume sales and inventory currently. The client explained that sales are recorded manually in an exercise book, while stock quantities are estimated by physically checking the shelves whenever new customers arrive. This sometimes results in perfumes being sold even though they have already run out.

The client also explained that calculating daily sales and remaining stock at the end of the day is time-consuming because all records are counted manually. During busy periods, some sales are forgotten or written down later, increasing the chances of mistakes.

When asked what exactly he expected from the proposed ScentHub system, the client said he wanted a simple system that would automatically reduce stock after each sale, allow him to quickly check available quantities, and generate sales records without requiring manual calculations. He also wanted only authorised workers to have access to the system because sales information is confidential.

The client emphasized that the system should prevent staff from recording sales when sufficient stock is unavailable and should display a clear error message explaining the problem instead of allowing incorrect records to be saved.

Requirement Transformation

Raw Client Statement

“Sometimes we sell perfumes and later realise the stock finished already because everything is written in books. I want the system to reduce the stock immediately after every sale.” -Client (Yussif Kangre Abdullai)

Functional Requirement

The system shall automatically reduce the inventory quantity of a perfume immediately after a successful sale has been recorded.

Non-Functional Requirement

The inventory update shall be completed within **2 seconds** after a successful sale and the stock information displayed to the user shall always reflect the latest available quantity.

User Story

As a shop attendant;

I want the system to automatically update perfume stock whenever I record a sale, so that I always know the correct quantity remaining and avoid selling unavailable products.

Acceptance Criteria

1. When a sale is successfully recorded, the product stock is automatically reduced.
2. The updated stock quantity is immediately displayed on the screen.
3. If the requested quantity exceeds available stock, the sale is rejected.
4. An error message clearly explains that there is insufficient stock.
5. No inventory record is changed if the transaction fails.

API / Interface Implication

The system should provide an endpoint such as:

POST /api/sales

The endpoint should:

- receive the sale information,
- validate stock availability,
- record the sale,
- update inventory,
- return either a success confirmation or an appropriate error message.

Data Implication

Each sale record should store:

- Sale ID
- Product ID
- Perfume name
- Quantity sold
- Selling price
- Date and time
- Sales attendant
- Remaining stock after the transaction

Test Idea

Attempt to sell 8 bottles of a perfume when only 5 bottles remain.

Expected Result:

- Sale is rejected.
- Stock remains at 5.
- Error message: "Insufficient stock available."

Open Question

Should managers be allowed to override stock shortages in special situations, or should the system always prevent sales when stock is unavailable?

Reflection

Before speaking with the client, I thought the main purpose of ScentHub was simply to replace the exercise books used to record sales. I believed that once sales and inventory were stored on a computer, most of the client's problems would be solved. However, after the discussion, I realised that understanding the client's daily work is just as important as building the software itself.

One thing that changed my thinking was how often small mistakes happen during busy business hours. The client explained that when there are many customers at the same time, it is easy to forget to write down a sale immediately or to make mistakes when updating the stock manually. This showed me that the system should not only store information but should also help prevent these mistakes from happening.

An assumption I had before the interview became false. I assumed that the biggest problem was keeping sales records, but the client made it clear that knowing the correct stock level at any time is even more important. If the stock information is not accurate, the business can disappoint customers by selling perfumes that are no longer available.

This conversation also helped me understand why software engineers spend time talking to clients before designing a system. What developers think users need is not always the same as what users need. Listening carefully helped me identify requirements that I would not have considered, such as automatically updating stock after every sale and preventing staff from recording sales when there is not enough stock.

Overall, this interview improved my understanding of requirements gathering. It showed me that good software starts with understanding the real problems of the people who will use it, instead of making assumptions about how their business works.